# Chapter 1: Distributed Consensus Protocols

## 1.1 Two-Phase Commit Protocol (2PC)

The Two-Phase Commit Protocol is an atomic commitment protocol for distributed databases.

Phase 1 (Prepare Phase): The coordinator node sends a PREPARE message to all cohort participants.

Phase 2 (Commit Phase): If all cohorts reply YES, the coordinator sends a GLOBAL-COMMIT command. If any cohort replies NO or times out, the coordinator sends a GLOBAL-ABORT command.

A key limitation of Two-Phase Commit Protocol is that it is a blocking protocol if the coordinator crashes during the commit phase.